

CS 499 Module One Assignment Template
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I. Self-Introduction: Address all of the following questions to introduce yourself.

A. How long have you been in the Computer Science program?

Hello! My name is Claudio. I am 47 years old, originally from Italy, and have been living in the United States for the past 10 years. I was accepted into SNHU's BS in Software Engineering program in March 2025, and I have accelerated my path to graduation by transferring slightly more than half of the required credits, most of which I earned through external general education courses.

B. What have you learned while in the program? List three of the most important concepts or skills you have learned.

1. Discovering a passion for embedded systems.

Although I had a technical background and worked as a computer technician in Italy for many years, my experience at SNHU helped me realize that I enjoy developing embedded devices much more than repairing computers and other electronic equipment. Designing and programming hardware-based systems has proven to be both more rewarding and more enjoyable for me.

2. Building a stronger foundation in programming.

I spent several years learning to program on my own, but I lacked a solid theoretical foundation and often viewed programming languages as more similar than they actually are. Through the program, I gained a better understanding of the different aspects of software development, including programming paradigms, algorithms, data structures, software design, and language-specific concepts. This experience helped me organize what I already knew and identify the areas I still need to develop to become a well-rounded software engineer.

3. Understanding the importance of software quality and security.

Before entering the program, I mainly focused on making software work. Through my coursework, I learned that developing professional software involves much more than implementing features. I gained experience with software testing, code coverage, and secure coding practices, and I became familiar with common vulnerabilities and the tools used to identify them. These experiences taught me that reliability and security are essential aspects of software engineering and that building quality software requires careful testing and attention to potential risks from the very beginning.

C. Discuss the specific skills you aim to demonstrate through your enhancements to reach each of the course outcomes.

For my capstone project, I identified my final project from CS 340 as a strong candidate to demonstrate my skills in software engineering, algorithms, data structures, and databases. The original artifact consisted of a Dash dashboard running inside JupyterLab and connected to a MongoDB database, but I plan to significantly enhance it across all three categories to showcase the knowledge and skills I have acquired throughout the program.

From a software engineering perspective, I intend to transform the original application into a full-stack MEAN application with a modular architecture, a two-tier design separating an authenticated internal dashboard from a public-facing web portal. In the area of algorithms and data structures, I plan to implement compound indexes on the most frequently queried fields, replace client-side pandas processing with server-side MongoDB aggregation pipelines, and introduce an LRU cache using a JavaScript Map data structure to avoid redundant database queries. For the database enhancement, I plan to expand the MongoDB backend by adding JSON Schema validation at the collection level, role-based access control with scoped user permissions, and a second collection to record rescue/training evaluation activity with a one-to-many relationship to the existing animals collection.

Throughout the enhancement process, I also intend to incorporate secure coding practices, including input validation, JWT authentication, and least-privilege access controls. In addition, I will document my design decisions and provide code reviews and narratives that clearly communicate technical concepts to both technical and non-technical audiences, demonstrating the communication and collaboration skills expected of a software engineer.

D. How do the specific skills you will demonstrate align with your career plans related to your degree?

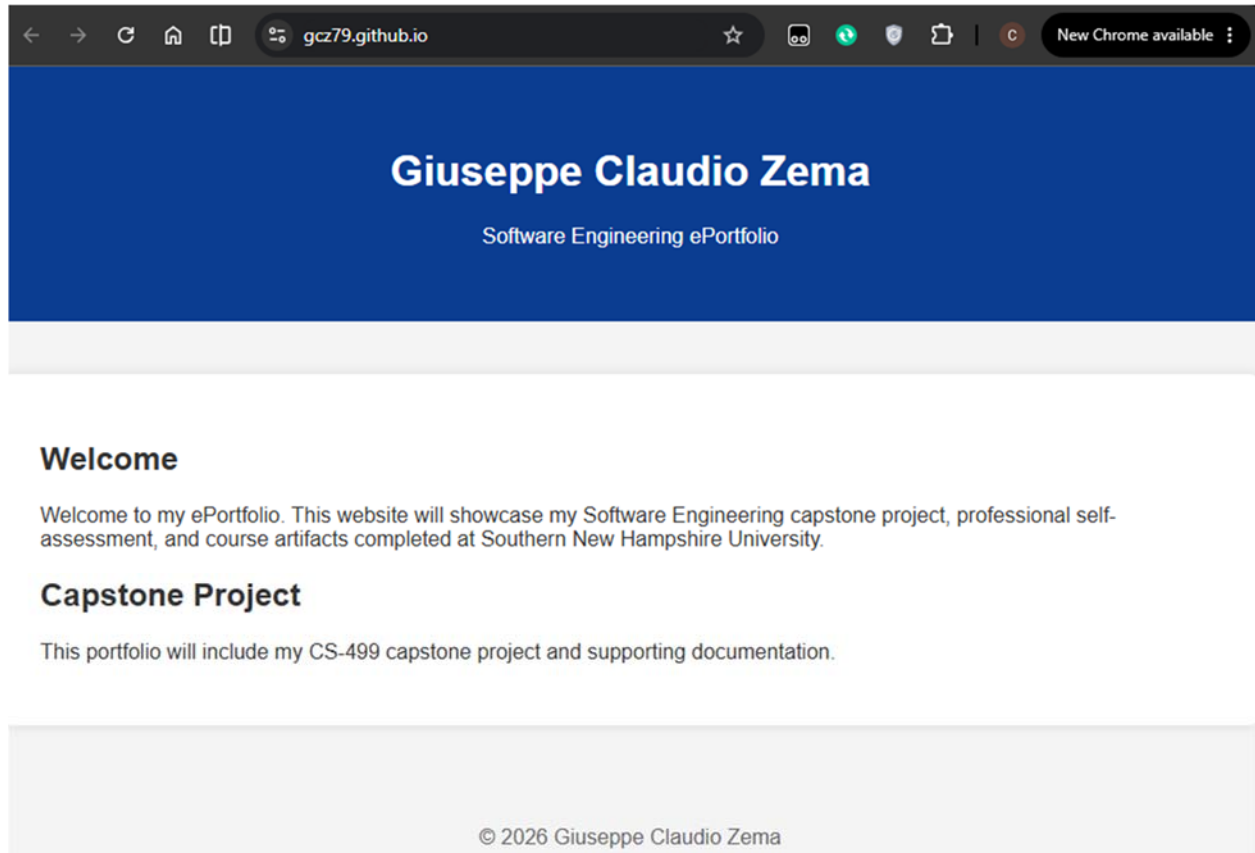
Although I have not yet committed to a specific career path within computer science, I believe the skills demonstrated through my enhancements will provide a strong foundation for a variety of software engineering roles. By expanding a previous project into a full-stack MEAN application, improving algorithms and search capabilities, and incorporating secure coding practices, I will strengthen my experience in software design, databases, and application development. These are transferable skills that will help me adapt to different opportunities as I continue to explore the field and determine the area that best fits my interests and strengths.

E. How does this contribute to the specialization you are targeting for your career?

At this point, I have not chosen a specific specialization within computer science. Throughout the program, I have had the opportunity to gain experience in many areas, including software engineering, databases, cybersecurity, testing, computer graphics, and embedded systems. As a result, I consider versatility to be one of my strengths. The enhancements planned for my capstone project will allow me to deepen my knowledge of full-stack development while continuing to build a broad technical foundation. This experience will help me make a more informed decision about a future specialization as I gain additional experience and enter the profession.

II. ePortfolio Set Up:

<https://gcz79.github.io/>



III. Enhancement Plan:

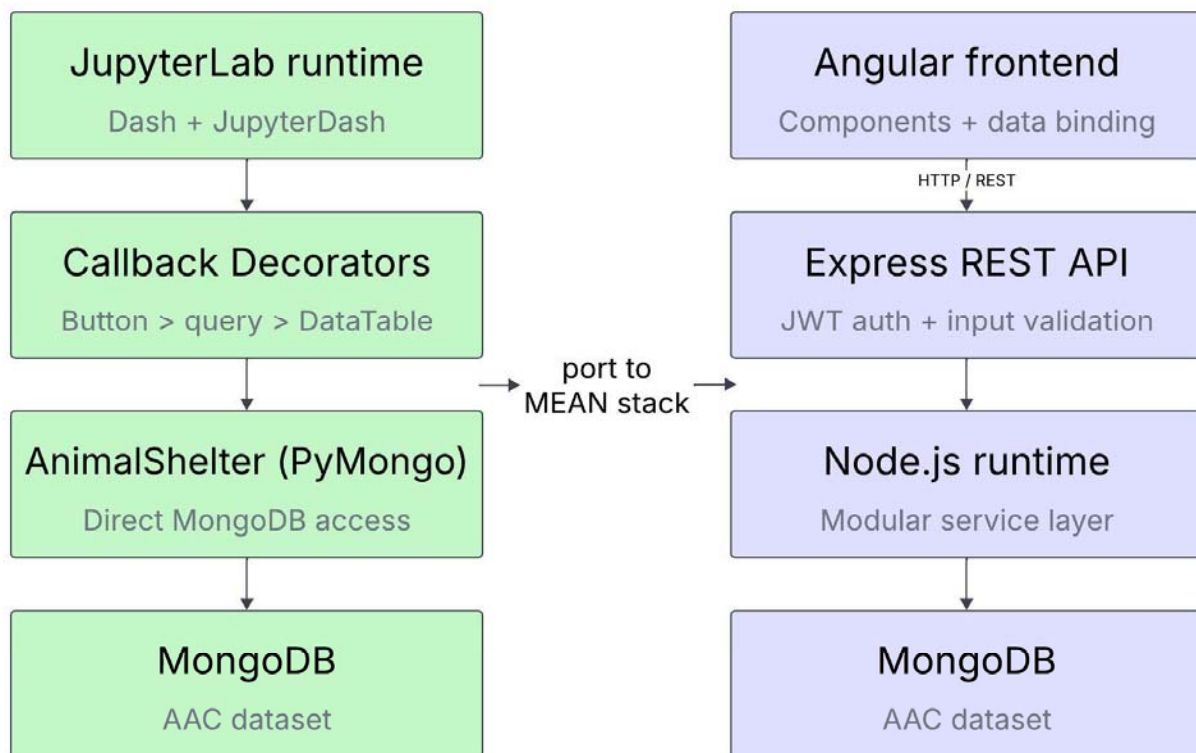
A. Category One: Software Engineering and Design

- i. Select an artifact that is aligned with the software engineering and design category and explain its origin. Submit a file containing the code for the artifact you choose with your enhancement plan.

The artifact selected for this category is the CS 340 Project Two Dashboard and CRUD Python Module, created in December 2025 for CS 340: Advanced Programming Concepts. The original application consists of a Dash dashboard running inside JupyterLab, connected to a MongoDB database containing records from the Austin Animal Center. It provides rescue/training-type filtering through preset buttons, a Leaflet map, a pie chart, column visibility controls, and a dataset cardinality summary, all driven by a Python CRUD module that wraps PyMongo.

- ii. **Describe** a practical, well-illustrated **plan** for enhancement in alignment with the category, including a pseudocode or flowchart that illustrates the planned enhancement.

The core transformation is a full-stack port from the Python/Dash/JupyterLab environment to a MEAN stack application (MongoDB, Express.js, Angular, Node.js). The Jupyter-based runtime and Dash component model will be replaced entirely by a production-style architecture: a Node.js/Express REST API serving data from the existing MongoDB database, and an Angular frontend that replicates and extends the original dashboard's functionality with a modular, component-based structure. In addition to the administrative dashboard, I plan to develop a public-facing website that presents the organization's mission, services, and contact module, providing visitors with information about the company and improving accessibility for potential adopters and other stakeholders.



What changes vs. what carries over



Angular frontend

HeaderComponent – Navigation and user status
FilterBarComponent – Rescue type filters
DataTableComponent – Search results table
MapComponent – Animal locations
ChartComponent – Statistics visualization

ColumnToggleComponent – Column selection
RxJS BehaviorSubject – Shared application state
AnimalService – REST API access
AuthService – Authentication

iii. Explain how the planned enhancement will demonstrate specific skills and align with course outcomes.

a. Identify and describe the specific skills you will demonstrate that align with the course outcome.

- Full-stack JavaScript/TypeScript development using the MEAN stack
- RESTful API design with Express.js, including route structure and middleware
- Angular component architecture, module design, and reactive state management with RxJS
- JWT-based authentication and authorization
- Two-tier application design separating public and authenticated access
- Input validation and server-side security controls
- Translating an existing application's logic to a new language and framework

b. Select one or more of the course outcomes below that your enhancement will align with.

- Outcome 1 - The migration from JupyterLab to a web-based application is fundamentally a decision about who can access the system. The original dashboard required a configured Python environment and familiarity with Jupyter, limiting its use entirely to technical users. The enhanced application introduces two audience tiers: an authenticated internal dashboard for shelter staff and coordinators to perform rescue/training filtering and data analysis, and a public-facing web portal accessible to anyone with a browser (including volunteers, donors, adopters, and partner organizations). This two-tier architecture directly supports building a collaborative environment that enables diverse audiences to engage with the organization's data and participate in its decision-making processes.
- Outcome 2 - The port requires designing professional-quality technical communications in a new context: the REST API is self-documenting through consistent route naming, and the Angular component structure makes the application's architecture readable and maintainable by collaborators.
- Outcome 4 - Migrating to the MEAN stack demonstrates the use of well-founded, industry-standard tools to deliver a production-capable solution that replaces a prototype Jupyter environment.
- Outcome 5 - Adding JWT authentication and input validation directly addresses the security gap in the original application, which stored credentials in plaintext and had no access controls.

B. Category Two: Algorithms and Data Structures

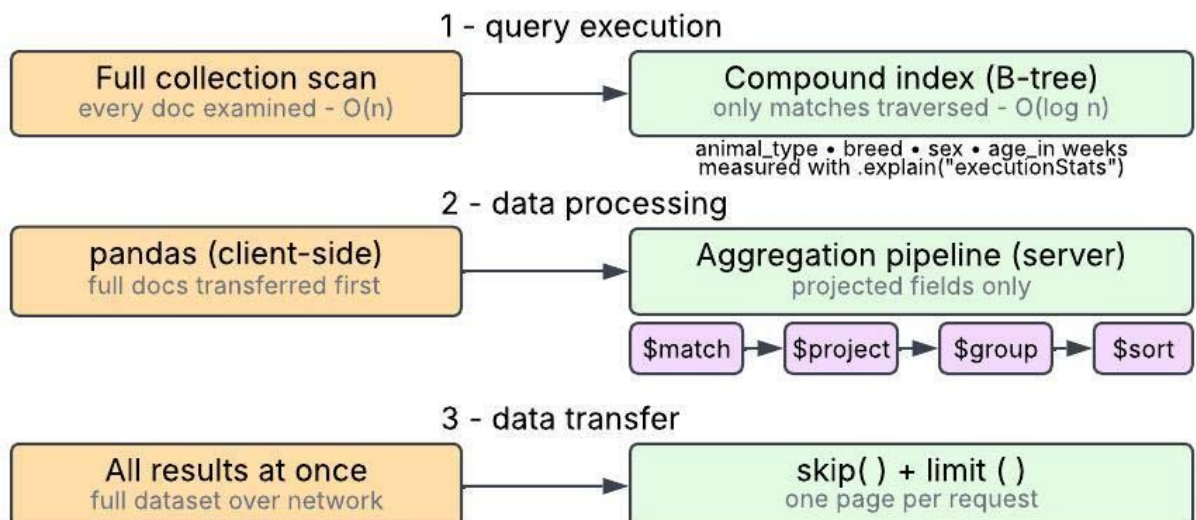
- i. **Select an artifact that is aligned with the algorithms and data structures category and explain its origin. Submit a file containing the code for the artifact you choose with your enhancement plan.**

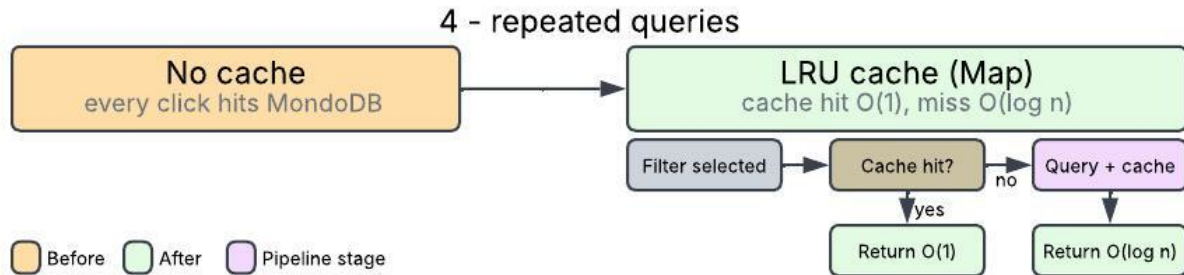
The artifact selected for this category is the same CS 340 CRUD module and dashboard created in December 2025. The current implementation retrieves data from MongoDB using basic `find()` queries with regex filters and processes results entirely in Python using pandas. Every query performs a full collection scan - $O(n)$ - regardless of how many documents match the filter. Breed distribution grouping for the pie chart is also computed client-side after the full result set has been transferred over the network. There is no indexing strategy, no server-side aggregation, and no caching of any kind, meaning every repeated filter selection re-executes the same database query from scratch.

- ii. **Describe a practical, well-illustrated plan for enhancement in alignment with the category, including a pseudocode or flowchart that illustrates the planned enhancement.**

The enhancement introduces three-layered algorithmic improvements to address the inefficiencies described above:

- Compound indexes on the four most-queried fields (`animal_type`, `breed`, `sex_upon_outcome`, `age_upon_outcome_in_weeks`), replacing full collection scans with B-tree-assisted lookups that significantly reduce the need to examine unrelated documents. Performance before and after will be measured using `.explain("executionStats")` and documented with a complexity analysis.
- Server-side aggregation pipelines using MongoDB's `$match`, `$project`, `$group`, and `$sort` stages, replacing the client-side pandas processing and reducing network transfer to projected fields only.
- Server-side pagination using MongoDB's `skip()` and `limit()` operators, returning data in fixed-size pages rather than transferring the full result set on every query. The trade-off between skip/limit and cursor-based pagination is documented in the complexity analysis.
- LRU cache implemented with a JavaScript Map data structure, storing results for recently used rescue/training filters. The Map is chosen specifically because it preserves insertion order, enabling $O(1)$ eviction of the least recently used entry without a separate tracking structure. Repeated filter selections return cached results in $O(1)$ rather than re-querying the database.





iii. Explain how the planned enhancement will demonstrate specific skills and align with course outcomes.

a. Identify and describe the specific skills you will demonstrate to align with the course outcome.

- B-tree index structure and compound index design in MongoDB
- Aggregation pipeline construction using `$match`, `$project`, `$group`, and `$sort`
- LRU cache implementation using a Map data structure with $O(1)$ hit and eviction
- Query performance measurement and analysis using `.explain("executionStats")`
- Big-O complexity analysis across multiple algorithmic components
- Server-side pagination design using skip/limit with documented trade-off against cursor-based alternatives
- Documented tradeoff reasoning: read vs. write cost, memory vs. latency, server-side vs. client-side processing
- Deliberate data structure selection: Map chosen over plain object for its insertion-order property enabling $O(1)$ eviction

b. Select one or more of the course outcomes listed under Category One that your enhancement will align with.

- Outcome 3 - This enhancement is built around algorithmic reasoning at multiple levels. The index design addresses query time complexity and is validated through empirical measurements rather than theoretical assumptions alone. The LRU cache introduces an explicit data structure, a Map, selected because its insertion-order property enables $O(1)$ eviction without auxiliary tracking. This non-trivial design decision requires understanding how the structure works internally. The aggregation pipeline shifts computational work to the appropriate layer, reducing network transfer. Each decision involves evaluating tradeoffs (read speed vs. write overhead, memory usage vs. query latency, server-side vs. client-side processing) which is what outcome 3 asks for.
- Outcome 4 - Using MongoDB's native aggregation framework, compound indexes, and a Map-based LRU cache reflects the use of well-founded, industry-standard techniques for scaling read-heavy dashboards. These are the same patterns used in production systems, not academic exercises.

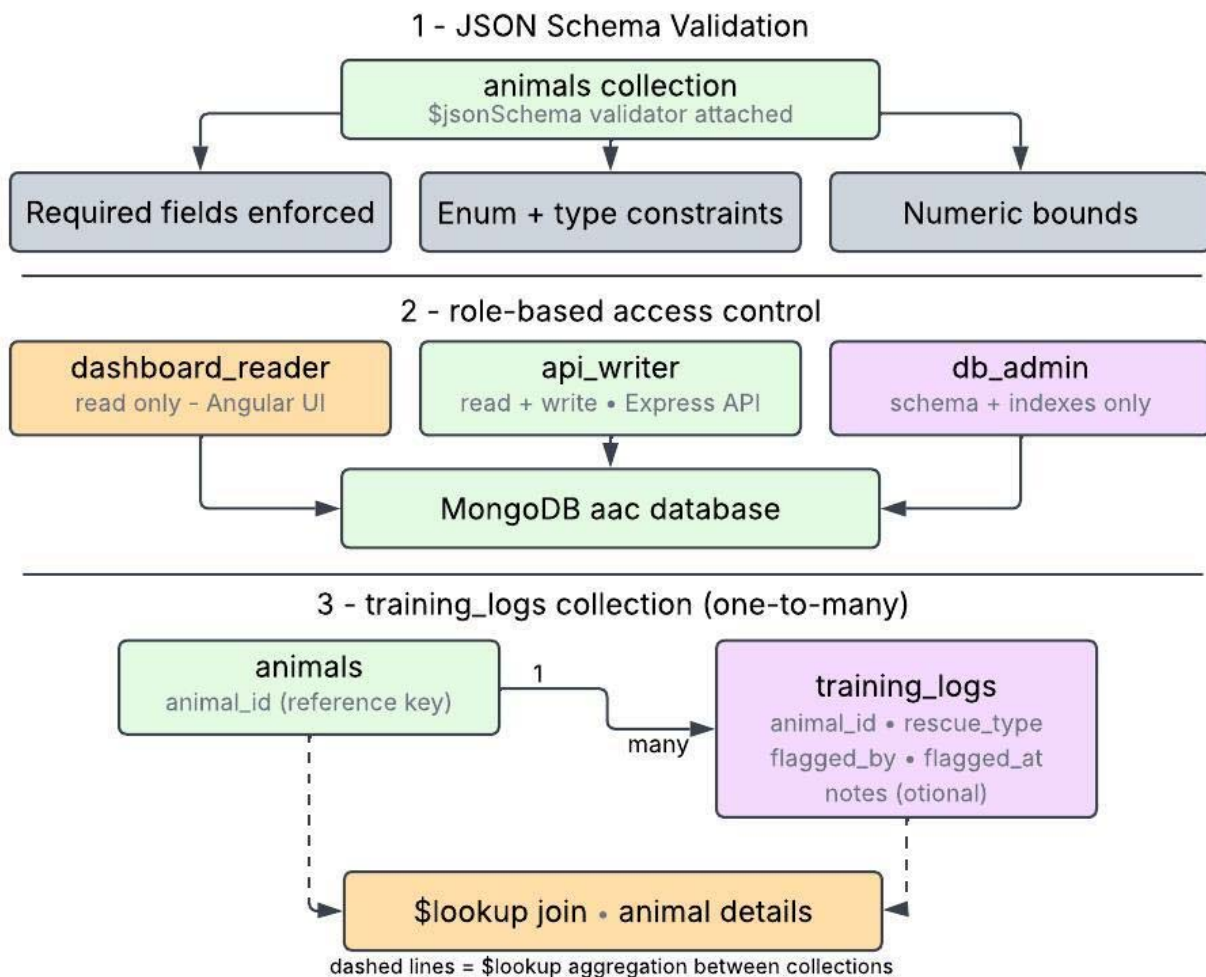
C. Category Three: Databases

- i. Select an artifact that is aligned with the database category and explain its origin. Submit a file containing the code for the artifact you choose with your enhancement plan. You may choose work from the courses listed under Category One.

The artifact selected for this category is the same CS 340 CRUD module and dashboard created in December 2025. The original MongoDB setup is minimal: a single collection with no schema validation, no indexing, hardcoded credentials, and a basic CRUD interface with no access controls or structured data management beyond what Dash requires.

- ii. Describe a practical, well-illustrated plan for enhancement in alignment with the category, including a pseudocode or flowchart that illustrates the planned enhancement.

The database enhancement expands the MongoDB backend with three additions: JSON Schema validation at the collection level to enforce data integrity, role-based access control (RBAC) using MongoDB's built-in user roles, and a second collection (`training_logs`) with a one-to-many relationship that records which animals were flagged for rescue/training evaluation and when. This introduces a relational-style data model within MongoDB and demonstrates schema design, referential structure, and database-level security.



iii. Explain how the planned enhancement will demonstrate specific skills and align with course outcomes.

a. Identify and describe the specific skills you will demonstrate that align with the course outcome.

- MongoDB JSON Schema validation and collection-level constraint design
- Role-based access control configuration with MongoDB user management
- Multi-collection schema design with a one-to-many document relationship
- \$lookup aggregation for cross-collection joins
- Database-level security as a complement to application-level security

b. Select one or more of the course outcomes listed under Category One that your enhancement will align with.

- Outcome 3 - Designing a second collection and a relational structure within a document database requires evaluating tradeoffs: embedding vs. referencing, schema flexibility vs. validation strictness. These are explicit design decisions with documented reasoning.
- Outcome 4 - Implementing schema validation and RBAC uses well-founded MongoDB features to improve the reliability and correctness of the database layer.
- Outcome 5 - RBAC directly addresses a security gap in the original application, which used a single user with full access and stored credentials in plaintext. Separating read-only and read-write roles and limiting admin access reflect a least-privilege security mindset.

IV. ePortfolio Overall Skill Set

A. Accurately describe the skill set to be illustrated by the ePortfolio overall.

i. Skills and outcomes planned to be illustrated in the code review

The code review will walk through the original CS 340 artifact before any enhancements, identifying specific structural and security weaknesses: hardcoded credentials, no input validation, no authentication, tightly coupled Dash callbacks, and no schema enforcement or indexing on the database side. This will demonstrate outcome 2 through professional oral and visual communication adapted to a peer or manager audience, and outcome 5 through a security-focused analysis that identifies concrete vulnerabilities in the existing codebase. The review will also outline each planned enhancement with enough technical detail to show that the work is purposefully planned across all three categories.

ii. Skills and outcomes planned to be illustrated in the narratives

Each narrative will focus on a different area of growth throughout the project. The software engineering narrative will discuss the transition from a callback-based Python application to a component-based Angular frontend and what I learned about modular design, separation of concerns, and designing applications for different types of users. The algorithms and data structures narrative will focus on improving query performance, using a Map data structure to implement an LRU cache, and understanding how to place computational work in the most appropriate layer of the application. The database narrative will describe the decisions behind the schema design, the importance of enforcing data integrity at the database level, and how role-based access control helps implement the principle of least privilege. Together, these narratives will demonstrate growth and support all five course outcomes.

iii. Skills and outcomes planned to be illustrated in the professional self-assessment

The self-assessment will serve as an introduction to the ePortfolio by connecting the knowledge and skills developed throughout the Computer Science program with the enhancements completed for the capstone project. It will address the required topics of collaboration, communication with stakeholders, algorithms and data structures, software engineering, databases, and security using examples from coursework completed throughout the program. In addition, it will present the three enhancements as different aspects of a single project, demonstrating my ability to approach software systems as integrated solutions rather than as a collection of separate and unrelated skills.